Data Science Canvas

Project: Alzheimer's Disease Detection

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Problem Statement

Business Case & Value Added

Early detection of Alzheimer's disease enables timely intervention, reduces long-term care burdens, and improves patient outcomes.
Current diagnosis relies on expensive imaging and subjective assessments.
Value Added:
•Scalable, low-cost screening tool using clinical + cognitive game data
•High sensitivity reduces missed diagnoses
•Interpretability facilitates clinical adoption
•Compact model deployable in resource-limited settings

Data Landscape

Required data:
Demographics, lifestyle, vitals, Symptoms, functional and cognitive tests, Game-based cognitive metrics.
All data already available and validated. No additional external data necessary for this version.

Model Selection

Based on data characteristics (32 clinical + cognitive features, mixed types, non-linear relationships):
Tree-based ensembles (Random Forest, XGBoost) for interpretability + non-linearity
Neural Network evaluated for comparison but deprioritized due to lower performance
Final selection: XGBoost due to highest accuracy (95.12%), best sensitivity, and 1.2 MB size

Model Requirements

To obtain a clinically viable model:
High sensitivity (>90%) to minimize false negatives
Maintain strong specificity (>95%)
Interpretability via feature importance + SHAP
Robust preprocessing and no data leakage
Real-time inference on CPU hardware
Stable performance across demographic groups (low bias)

Software & Libraries

Python 3.10, Jupyter, PyTorch, scikit-learn, XGBoost, Pandas, NumPy, Matplotlib/Seaborn for EDA Streamlit for application UI SHAP for interpretability These libraries form a standard, production-ready solution.

Skills

Project requires team expertise across:
Clinical domain understanding of dementia indicators
Data engineering and preprocessing
Python, ML pipelines, scikit-learn, XGBoost, PyTorch
Hyperparameter tuning & model evaluation
Statistical analysis & hypothesis testing
SHAP for interpretability
Streamlit for UI development
Cognitive game design & signal mapping

Execution & Evaluation

Model Evaluation

Key indicators for quality control:
Accuracy, Precision, Recall, F1-score
ROC-AUC (≥ 0.94 achieved)
Confusion matrix analysis
test for statistical model comparison
Feature importance stability
Real-time monitoring required during deployment to identify drift or erroneous inputs.

Data Storytelling

Target audience: clinicians, researchers, and diagnostic staff
Clear explanation of risk tiers (Very Low → Very High)
SHAP plots showing per-patient reasoning
Comparisons of model performance
Visualizations: feature importance, distributions, correlation heatmaps
Dashboard for EDA transparency
Objective: communicate why the model predicts a diagnosis, not just what.

Data Collection & Preparation

Data Selection & Cleansing

Available data from 2,150 patients with 32 features:
No missing values or duplicates
Outlier inspection performed
Continuous features normalized using Standard Scaler
Categorical features validated
Cognitive game outputs mapped to structured metrics
Dataset is clean and directly usable after standardization.

Data Integration

All data sources clinical, behavioural, cognitive game metrics integrated into a unified feature matrix.

Processing system: Python pipeline with: Schema validation
Preprocessing
Model loading
Risk-tier mapping
Final model deployed within a Streamlit application.

Data Collection

Kaggle dataset
Clinical records: demographics, vitals, medical history
Standardized cognitive/functional assessments (MMSE, ADL, FA)
Self-reported symptoms
Four designed cognitive mini-games measuring memory, reasoning, reaction time, and task switching
Properties required:
Values must fall within clinically valid ranges
Cognitive test outputs must follow consistent mapping schema

Explorative Data Analysis

Key EDA insights:
Age differentiates groups strongly (A: 81.2 vs H: 72.1)
Cognitive + functional measures are the strongest predictors (MMSE, ADL, FA)
Lifestyle and medical history features show modest but meaningful interactions
Minimal multicollinearity → all 32 features retained
Strong symptom correlation clusters (memory, behavioral issues)
EDA validated that cognitive/functional decline is the primary signal for prediction.